

UNIT-1

LESSON 1 : Operating System – Basics

1. Operating System (OS)

An Operating System is system software that acts as an interface between user and hardware.

Main Goal

Efficient use of hardware

Convenience to user

2. Functions of Operating System

Process management

Memory management

File system management

I/O device management

Security & protection

Resource allocation

LESSON 2 : Evolution of Operating System

1. Early Systems

No OS (Bare machine)

One program at a time

2. Batch Operating System

Jobs collected and executed in batches

No user interaction

Advantages

Better CPU utilization

Disadvantages

Long turnaround time

3. Multiprogramming Systems

Multiple programs in memory

CPU switches between them

4. Time Sharing Systems

CPU time divided into time slices

Supports interactive users

5. Modern Operating Systems

GUI based

Multitasking, networking, real-time support

LESSON 3 : Classification of Operating Systems

1. Batch Operating System

Processes executed in batches

Example: Early IBM systems

2. Interactive Operating System

Direct user interaction

Fast response time

3. Time Sharing Operating System

Multiple users share CPU

Example: UNIX

4. Multiprocessor System

More than one CPU

Increased throughput

5. Multiuser System

Multiple users at same time

Example: Linux server

6. Multiprocess System

Multiple processes running simultaneously

7. Real-Time Operating System

Responds within strict time limits

Types

Hard real-time

Soft real-time

8. Multithreaded System

Supports multiple threads in a process

Better responsiveness

LESSON 4 : Operating System Structure

1. Layered Structure

OS divided into layers

Each layer uses services of lower layer

Advantages

Easy debugging

Modular design

Disadvantages

Performance overhead

2. Monolithic Kernel

Entire OS in single kernel space

Advantages

High performance

Disadvantages

Difficult to maintain

3. Microkernel System

Minimal kernel

Services run in user space

Advantages

Better security

Easy extensibility

Disadvantages

Communication overhead

LESSON 5 : Process Concept

1. Process

A process is a program in execution.

2. Process States

New

Ready

Running

Waiting

Terminated

3. Process Transition Diagram

Shows movement of process between states.

Transitions

New → Ready

Ready → Running

Running → Waiting

Running → Terminated

LESSON 6 : Process Control Block (PCB)

PCB

Data structure used by OS to store process information.

Contains

Process ID

Process state

Program counter

CPU registers

Memory management info

I/O status

LESSON 7 : Process Address Space

Process Address Space

Logical memory allocated to a process.

Components

Code segment

Data segment

Heap

Stack

LESSON 8 : Process Identification Information

Includes:

Process ID (PID)

Parent Process ID

User ID

Group ID

Used for tracking and security.

LESSON 9 : Threads

Thread

Smallest unit of CPU execution within a process.

Advantages of Threads

Faster context switching

Resource sharing

Better responsiveness

Thread Management

Creation

Scheduling

Synchronization

Termination